

• OFFICIAL ENGINEERING DOCUMENTATION • V2.6.4

VORTEX PT

Comprehensive User Manual & Technical Detailing Guide for Automated Post-Tensioned Concrete Drafting, Parabolic Profiling, and AutoCAD DWG Generation.



PRIMARY ENGINE

RAM Concept & ADAPT
(.adm / .inb)

OUTPUT STANDARDS

AutoCAD DWG / DXF /
Excel

COMPLIANCE CODE

ACI 318 / PTI Manual

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1. Introduction & Core Architecture

Vortex PT is a specialized desktop engineering suite designed to bridge the workflow gap between finite-element post-tensioned design software (**Bentley RAM Concept** and **ADAPT-Builder**) and construction-ready shop drawings in **AutoCAD (.dwg / .dxf)**.

Traditional shop drawing workflows require draftsmen to manually trace tendon spans, hand-calculate chair support heights, transcribe profile points, and manually draw anchor setbacks. Vortex PT eliminates these manual steps, reducing a multi-day drafting task to under 30 seconds while guaranteeing zero transcription errors.

TARGET USERS Structural Engineers, PT Contractors, CAD Detailers	EXECUTION ENVIRONMENT 100% Offline Desktop, Windows 10 / 11 64-bit
INPUT FORMATS SUPPORTED Bentley RAM Concept (*.cpt, *.cpt8) & ADAPT-Builder (*.adm with .inb/.inp)	OUTPUT FORMATS GENERATED Layered AutoCAD DWG, DXF, Microsoft Excel (.xlsx), CSV

i

No Host Software License Required

Vortex PT features autonomous native binary parsers. You do **not** need a Bentley RAM Concept or ADAPT-Builder license running on your drafting or site workstations to open models and produce fabrication drawings.

02 Opening & Importing Models

2.1 Supported File Types

Vortex PT natively imports structural data directly from the industry's two dominant post-tensioned design platforms:

- **Bentley RAM Concept Files (*.cpt, *.cpt8):** Direct parsing of proprietary binary archives including slabs, drop panels, openings, columns, shear walls, beams, latitude tendons, longitude tendons, and control points.
- **ADAPT-Builder Project Models (*.adm with companion *.inb / *.inp):** Direct loading of ADAPT Floor Pro, MAT, and Edge models. You select the native .adm file directly in Vortex PT, which automatically reads the companion .inb / .inp file of the same name.

2.2 ADAPT-Builder Import Architecture (.adm + .inb)

In ADAPT-Builder, models are saved with the default extension **.adm**. To import an ADAPT model into Vortex PT:

- Vortex PT directly opens the **.adm** file from the file open dialog.
- **Mandatory Pairing Rule:** Beside your .adm file in the same folder, there must exist a companion Generic Model file (**.inb** or **.inp**) bearing the **exact same base name** (e.g. Building_Level12.adm paired with Building_Level12.inb or Building_Level12.inp).
- Vortex PT also automatically checks companion project subfolders (<model_folder>/<model_name>/) and ADAPT automatic backup directories (backup/).



How to Export from ADAPT-Builder

1. Open your project in **ADAPT-Builder** (Floor Pro, MAT, or Edge).
2. Go to menu: **File** → **Export** → **Generic Model (*.inp / *.inb)**.
3. Save the exported file in the same directory as your .adm file with the matching name.
4. In Vortex PT, click **Open Model** and select the .adm file. Vortex PT reads the paired data and instantly renders the entire floor plate with 100% geometric fidelity!

2.3 Step-by-Step Model Loading Workflow

1. Click the vibrant electric cyan **" + Open Model "** button located at the top left of the workspace toolbar, or press the universal shortcut **Ctrl + O**.
2. In the Windows File Dialog, browse to your model directory and select your **Bentley RAM Concept (.cpt / .cpt8)** or **ADAPT-Builder (.adm)** file.
3. Vortex PT initiates its multi-stage extraction pipeline:
 - Reads concrete geometry: slab perimeters, thickness zones, drop caps, and core openings.
 - Extracts structural supports: columns, shear walls, and perimeter foundation lines.

- Assembles continuous tendon strings from segmental FEA elements, calculating curvature, inflection transitions, and anchorage conditions.
4. Upon completion, the main CAD viewport renders the complete slab floor plate, and the **Project Metrics Card** populates immediately.



Figure 2.1: Live 2D CAD Workspace after opening B01-PART 2.cpt . Displays 70 continuous tendons, slab boundaries, drop panels, columns, the floating Project Metrics HUD, and Tendon Y-3 longitudinal profile.

03

2D CAD Viewport & Layer Controls

3.1 Viewport Navigation & Canvas Controls

The central viewport is powered by a high-performance vector rendering canvas designed for large multi-span commercial slabs:

ACTION	INPUT MECHANISM	DESCRIPTION
Pan Canvas	Middle Mouse Click & Drag or Right Mouse Click & Drag	Smoothly translates the viewport in any direction without altering zoom scale.
Zoom In / Out	Mouse Scroll Wheel	Continuously zooms centered directly onto the cursor coordinate.
Fit to Extents	Press Spacebar or click "Fit View"	Calculates the bounding box of all slabs and centers the entire floor plate.
Select Tendon	Left Mouse Click on Tendon	Selects the tendon, activates bright cyan vector highlight, and docks profile view.
Tendon Search	Ctrl + F → Type Tendon Mark	Instantly queries tendons (e.g. X-14, Y-3), highlights, and centers the view.

3.2 Layer Visibility Chips

Located in the command bar above the CAD canvas, interactive layer chips permit instant toggling of individual drafting layers:

- **Slabs (Gray #475569):** Concrete floor boundaries and thickness elevation zones.
- **Drops (Orange #F59E0B):** Drop panels and drop caps at column heads.
- **Columns (Yellow #FBBF24):** Rectangular, circular, and wall support sections.
- **Beams (Cyan #06B6D4):** Post-tensioned band beams and drop beams.
- **Tendons (Latitude Magenta #D946EF / Longitude Teal #14B8A6):** Tendon c/c lines and anchor bubbles.
- **Chairs & Dimensions:** Chair height callouts, remainder dimensions, and span spacing.

3.3 Floating Project Metrics Card

The upper-right HUD card provides real-time engineering summary statistics for the loaded structure:

- **Slab Area & Thickness:** Net structural floor area (m²) and primary plate depth (mm).
- **Structural Members:** Drop panels count and transfer/band beams detected.
- **Tendon & Strand Length:** Total active tendon count and linear strand length (m).
- **Strand Weight Takeoff:** Total fabricated strand weight (kg) calculated using your configured unit weight.
- **Anchorage Breakdown:** Exact count of Live Stressing Jacks vs. Dead Blind End Anchors.

04 Parabolic Tendon Profiling & Chairs

4.1 The Longitudinal Profile Viewer

Selecting any tendon triggers the bottom **Longitudinal Tendon Profile Dock**. This viewer displays the continuous cross-sectional elevation of the tendon from its starting anchor to its ending anchor.



Figure 4.1: Continuous 11-Span Longitudinal Elevation Profile of Tendon X-16. Features high support points over columns, low mid-span sags with fabricated chair heights, pan box setback blocks, and span dimension strings.

4.2 Mathematical Profile Modeling & Inflection Points

Vortex PT calculates true multi-span second-degree and third-degree reverse parabolic curves across each support bay:

- **High Support Points (Hi):** Peak elevations located over column centerlines or interior wall supports.
- **Low Sag Points (Li):** Maximum sag elevations located at mid-spans to optimize balancing load.
- **Inflection Transition Points (Xinf):** Mathematically computed points of zero curvature where the curve transitions smoothly between concave-up and concave-down parabolas.

4.3 Automatic Chair Support Calculations

Fabrication chairs are calculated at regular field intervals (default 1,000 mm) along the tendon:

**Chair Height Governing Formula**

Fabricated Chair Height = Tendon Centerline Profile (CGS) – Deduction Constant

Calculated chair heights are automatically rounded to your specified fabrication increment (e.g. 5 mm steps).

**Anchor Placement vs. Chairs**

Anchorage points (Live Jacking Ends and Dead Cast-In Anchors) are physical fittings affixed to the slab formwork. Vortex PT excludes anchors from the chair table summary to prevent accidental chair over-ordering on site.

05 Engineering Compliance & QA Alerts

5.1 Automated Quality Assurance Engine

Every imported model undergoes continuous geometric and structural integrity diagnostics based on **ACI 318-19** and the **Post-Tensioning Institute (PTI) Manual**. When an issue is detected, an amber "⚠ QA Warning" badge illuminates in the top command bar.

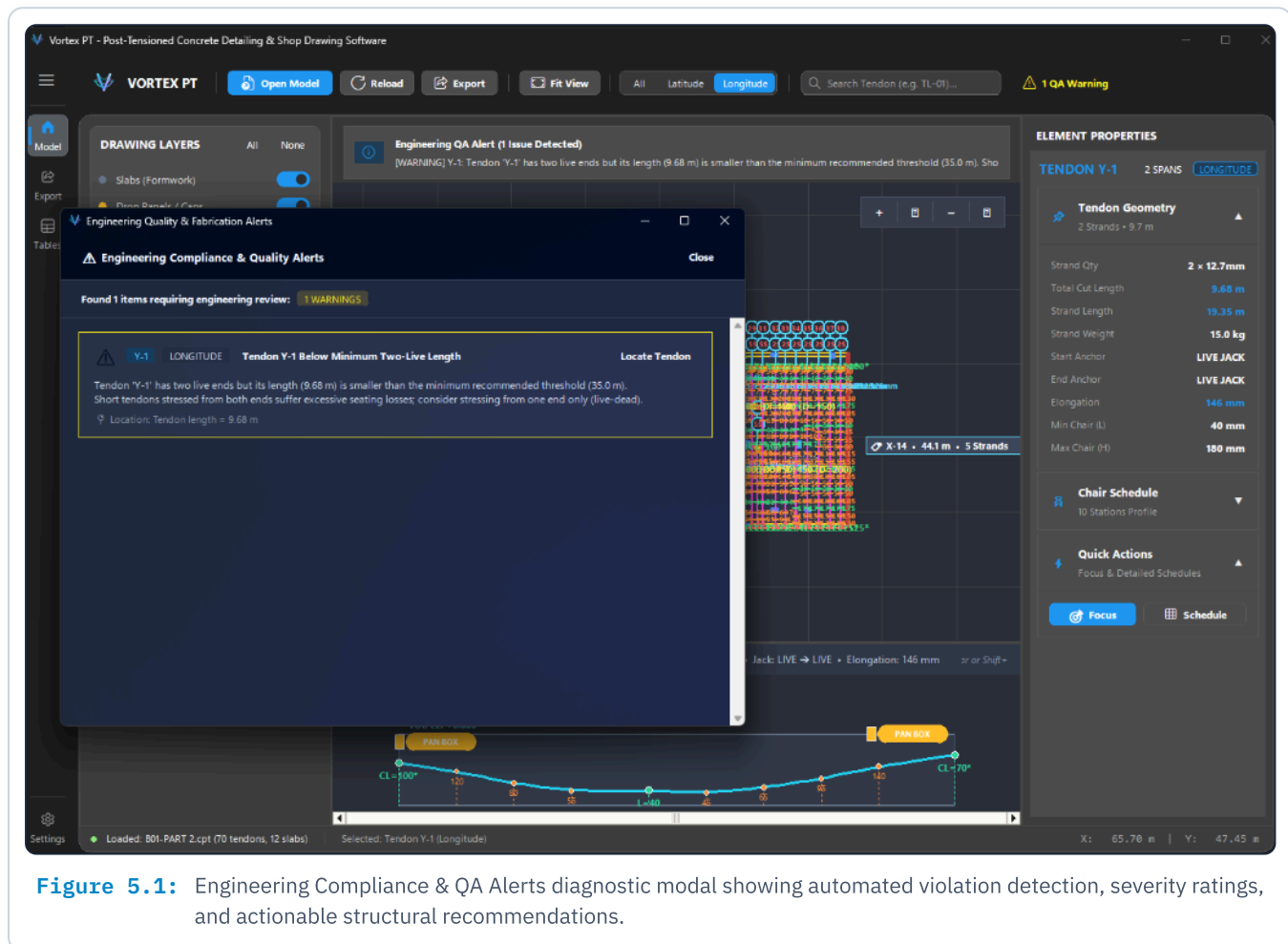


Figure 5.1: Engineering Compliance & QA Alerts diagnostic modal showing automated violation detection, severity ratings, and actionable structural recommendations.

5.2 Core Diagnostic Rules

A. Minimum Concrete Cover Verification

- **Top Cover Check:** Verifies that Slab Depth – Profile Elevation $\geq$ Min Cover (default: 25 mm).
- **Bottom Cover Check:** Verifies that Profile Elevation $\geq$ Min Cover (default: 25 mm).
- **Alert Condition:** If strand trajectory violates cover rules at high or low points, Vortex PT flags the exact bay and span coordinate.

B. Double Live End Minimum Length Threshold

- **Rule Rationale:** Short tendons configured with live stressing jacks at both ends lead to wedge seating loss issues and unnecessary labor costs.
- **Threshold Setting:** In **Settings**, you configure the warning threshold (e.g. 35.0 m). Tendons with two live ends shorter than this threshold trigger an alert advising a single live end configuration.

C. Two Dead Ends Critical Validation

- **Fatal Check:** Scans the model for any tendon terminated with dead-end (cast-in) anchors at both boundaries.
- **Severity:** High / Critical. Tendons cannot be stressed if both ends are encapsulated in blind anchors.

06

Detailing Settings & CAD Blocks

6.1 Windows 11 Fluent Settings Console

Click the **"Settings"** gear icon in the left navigation rail or press `Ctrl` + `,` to open the comprehensive configuration console.

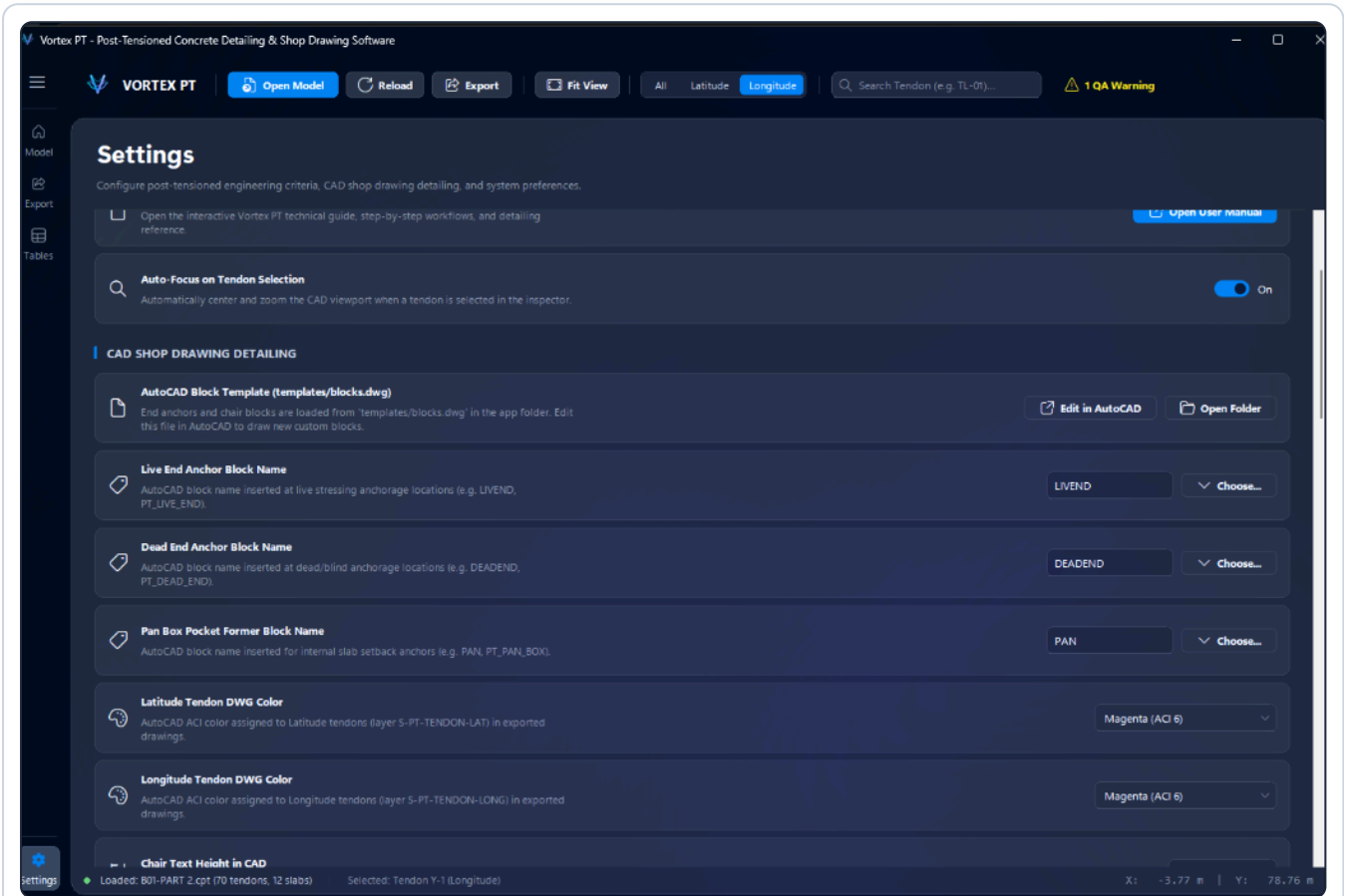


Figure 6.1: CAD Shop Drawing Detailing configuration cards, including template selection, block name bindings (LIVEND, DEADEND, PAN), ACI layer color overrides, and strand properties.

6.2 Custom AutoCAD End Blocks (DWG / DXF)

Vortex PT seamlessly imports standard and company-specific anchor blocks from your master template drawing located at `templates/blocks.dwg`.

SETTING CARD	DEFAULT BLOCK	ENGINEERING PURPOSE
Live End Block	LIVEND	Stressing pocket anchor inserted at slab edge live stressing jacks.
Dead End Block	DEADEND	Cast-in blind anchor bulb / onion anchor embedded in concrete edge.
Pan Box Block	PAN	Interior stressing blockout pan box inserted along continuous slab runs.

6.3 Customizing Your Own Blocks in AutoCAD

1. In **Settings** → **CAD Shop Drawing Detailing**, click **"Edit in AutoCAD"**.
2. AutoCAD will open `templates/blocks.dwg`.
3. Draw your firm's custom anchor geometries (oriented along the horizontal X-axis).
4. Save the file. In Vortex PT, click **"Choose..."** to select your newly created block name!

6.4 Material & Dimension Properties

- **Chair Text Height:** Configures the annotation height of chair labels in AutoCAD (default: 200 mm).
- **Strand Unit Weight:** Sets the specific weight per linear meter (default: 0.775 kg/m for 0.5" / 12.7 mm 7-wire strand). Used for all takeoff calculations.
- **Layer Color Overrides:** Assign specific AutoCAD ACI colors to Latitude (e.g. Magenta [6]) and Longitude (e.g. Cyan [4]) tendons for rapid field discrimination.

07 AutoCAD DWG/DXF Shop Drawing Export

7.1 Export Console Overview

Click the **"Export"** icon in the left navigation rail or press **Ctrl + E** to access the production drawing generator.

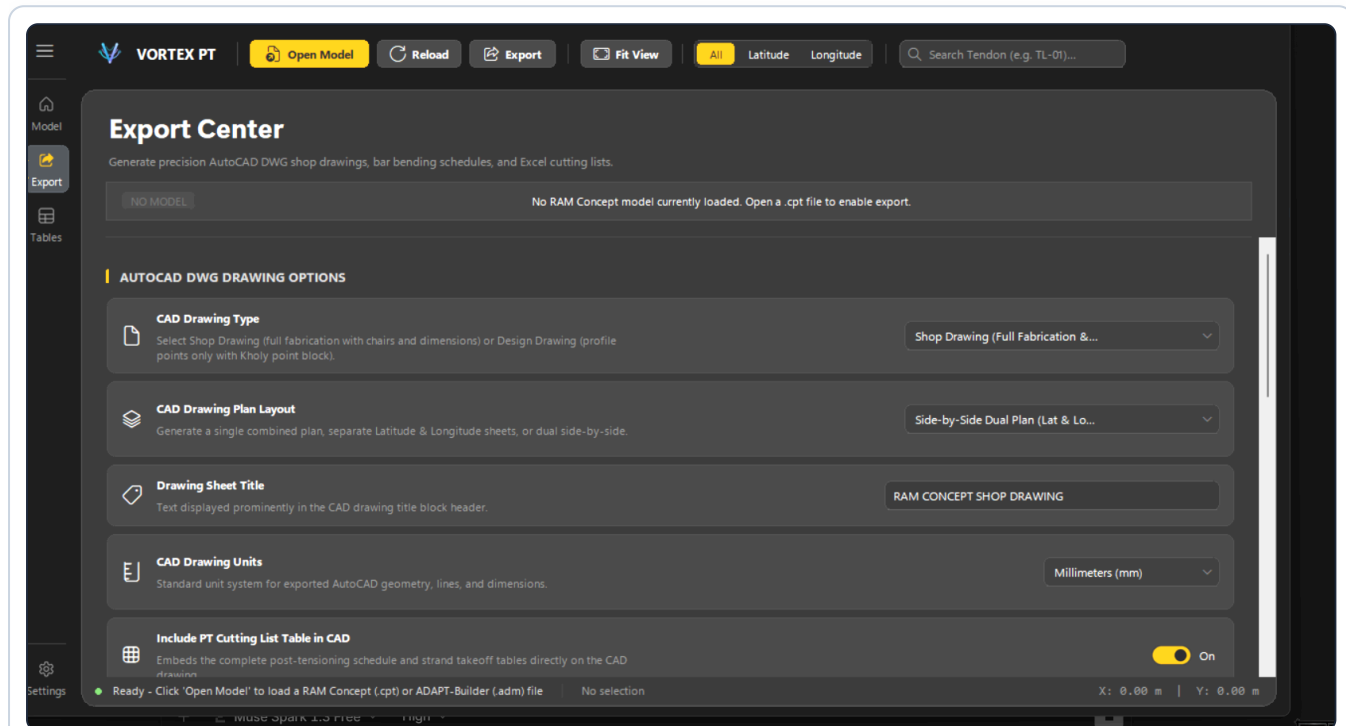


Figure 7.1: Production Drawing Export Console. Configure drawing types, layout styles, annotation layers, and direct AutoCAD DWG / DXF generation.

7.2 Production Drawing Configurations

A. CAD Drawing Type

- **Shop Drawing (Full Fabrication & Chairs):** Complete fabrication-level drawing including chair support tags, remainder dimensions, tendon IDs, elongation markers, and anchor blocks.
- **Design Drawing (Profile Points Only):** Streamlined consultant drawing with Kholy profile point blocks and high/low labels.

B. Sheet Layout Options

- **Side-by-Side Dual Plan [Recommended]:** Automatically offsets and places the Latitude drawing and Longitude drawing side-by-side on a single DWG layout for streamlined contractor printing.
- **Combined Plan:** Both Latitude and Longitude systems superimposed on a single floor plan.
- **2 Separate Plans:** Generates distinct independent files (e.g. Model_Lat.dwg and Model_Long.dwg).

7.3 Embedded Features

- **AutoCAD Layer Architecture:** All geometries are partitioned into dedicated layers (PT-TENDON-LAT, PT-TENDON-LONG, PT-CHAIRS, PT-ANCHORS, PT-DIMS) matching standard drafting protocols.
- **Embedded Schedule Table:** Optionally embeds the complete PT tendon schedule directly into model space.

08 Fabrication Schedules & Takeoffs

8.1 Schedules Workspace

Click the **"Schedules"** icon in the left navigation rail to access interactive material cutting schedules and bar chair summaries.

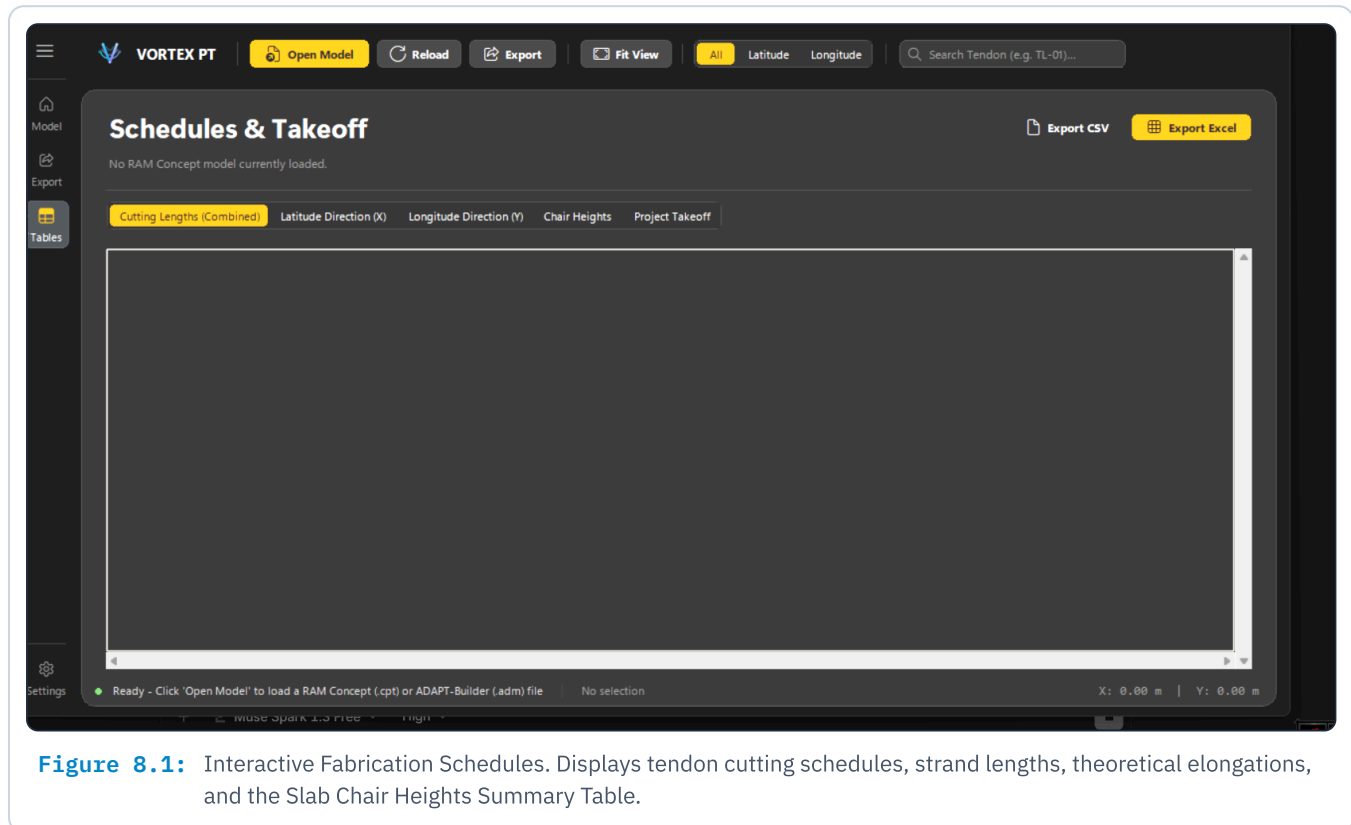


Figure 8.1: Interactive Fabrication Schedules. Displays tendon cutting schedules, strand lengths, theoretical elongations, and the Slab Chair Heights Summary Table.

8.2 Schedules Provided

1. **Combined Cutting Schedule:** Complete bill of materials listing Tendon Mark, Strand Count, Jacking Ends, Plan Length (m), Fabricated Cut Length (including stressing tail), Weight (kg), and Theoretical Elongation (mm).
2. **Directional Schedules (Latitude / Longitude):** Filtered schedules partitioned for specific installation crews.
3. **Slab Chair Heights Summary Table:** Groups all chairs across the entire floor plate by fabricated height (e.g. 25 mm, 30 mm, ..., 200 mm) with exact counts for Latitude and Longitude.

8.3 Exporting to Microsoft Excel & CSV

Click the **"Export Excel (.xlsx)"** button or press **Ctrl + Shift + E**. Vortex PT generates a multi-tab workbook with pre-formatted engineering headers, mathematical column formulas, and print-ready page setups.

09

Node-Locked Licensing & Activation

9.1 Hardware-Bound Security

Vortex PT operates 100% offline using high-grade Ed25519 digital signature cryptography. Each workstation is identified by a unique, tamper-proof **Machine ID** computed from hardware parameters.

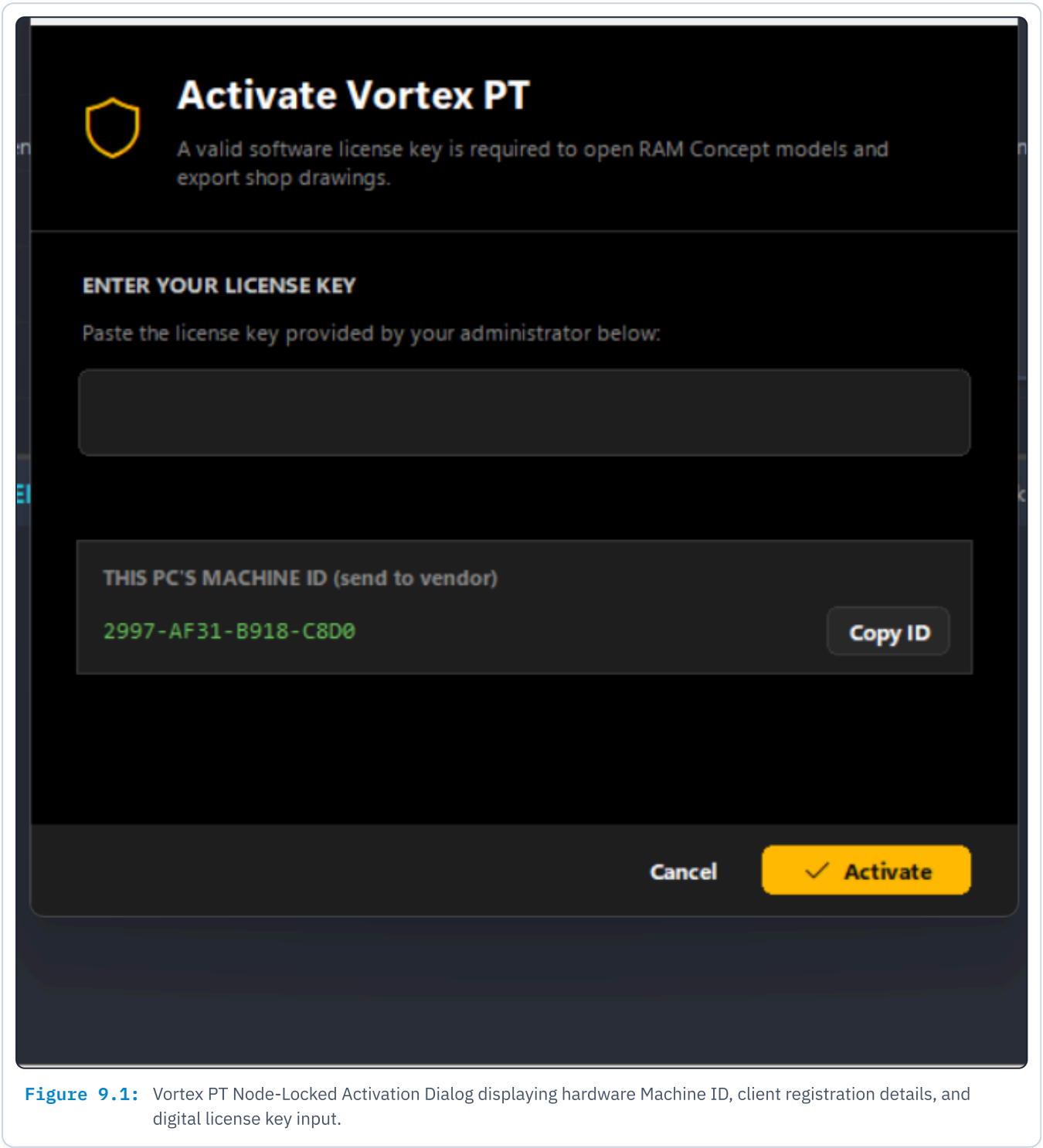


Figure 9.1: Vortex PT Node-Locked Activation Dialog displaying hardware Machine ID, client registration details, and digital license key input.

9.2 Activation Procedure

1. Upon first launch (or when clicking "**License Status**" in Settings), the Activation Dialog appears.
2. Click "**Copy Machine ID**" to copy your unique workstation identifier.
3. Send your Machine ID to your administrator or sales representative to receive your signed license key.
4. Paste the key into the "**License Key**" box and click "**Activate License**".
5. The software verifies the digital signature locally and activates instantly. No internet connection is ever required.

10 Keyboard Shortcuts & Engineering Reference

10.1 Complete Shortcut Reference

SHORTCUT	FUNCTION / COMMAND	CONTEXT
Ctrl + O	Open RAM Concept (.cpt) or ADAPT (.adm / .inb) Model	Global
Ctrl + E	Navigate directly to Export View	Global
Ctrl + Shift + E	Quick Export Schedules to Excel (.xlsx)	Global / Schedules
Ctrl + F	Focus Tendon Quick Search Field	Model View
Ctrl + ,	Open Settings Console	Global
Spacebar / F	Fit Entire Floor Plate to Viewport Extents	Model View
Esc	Clear Tendon Selection & Close Floating Panels	Model View
F5	Reload & Re-parse Current Model File	Global

10.2 Standard 0.5" (12.7 mm) Strand Technical Specifications

PROPERTY	VALUE (SI METRIC)	DESIGN STANDARD
Nominal Strand Diameter	12.7 mm (0.50 inch)	ASTM A416 / ISO 6934
Nominal Cross-Sectional Area (Aps)	98.7 mm ² (0.153 in ²)	Grade 270 (1860 MPa) Low Relaxation
Strand Unit Weight	0.775 kg/m (0.521 lb/ft)	Standard 7-Wire High Tensile Strand
Modulus of Elasticity (Ep)	195,000 MPa (28,500 ksi)	Theoretical Elongation Calculations
Minimum Breaking Strength (Ppu)	183.7 kN (41.3 kips)	ASTM A416 Tensile Strength
Initial Jacking Force (0.80 fpu)	147.0 kN	Standard PTI Field Stressing



Technical Support & Licensing Inquiries

For software inquiries, license key renewals, or technical support, visit the official portal at <https://vortex-pt.com> or email support@vortex-pt.com.